AFRICAST

TIDY TIME SERIES & FORECASTING IN R

# Basic modelling and forecasting




Session 4 of 10

# Your forecasting

What do you forecast (or wish you could) in your work?




poll.nectric.com.au/s/
africast-modelling

**21** responses

● CLOSED 5 connected

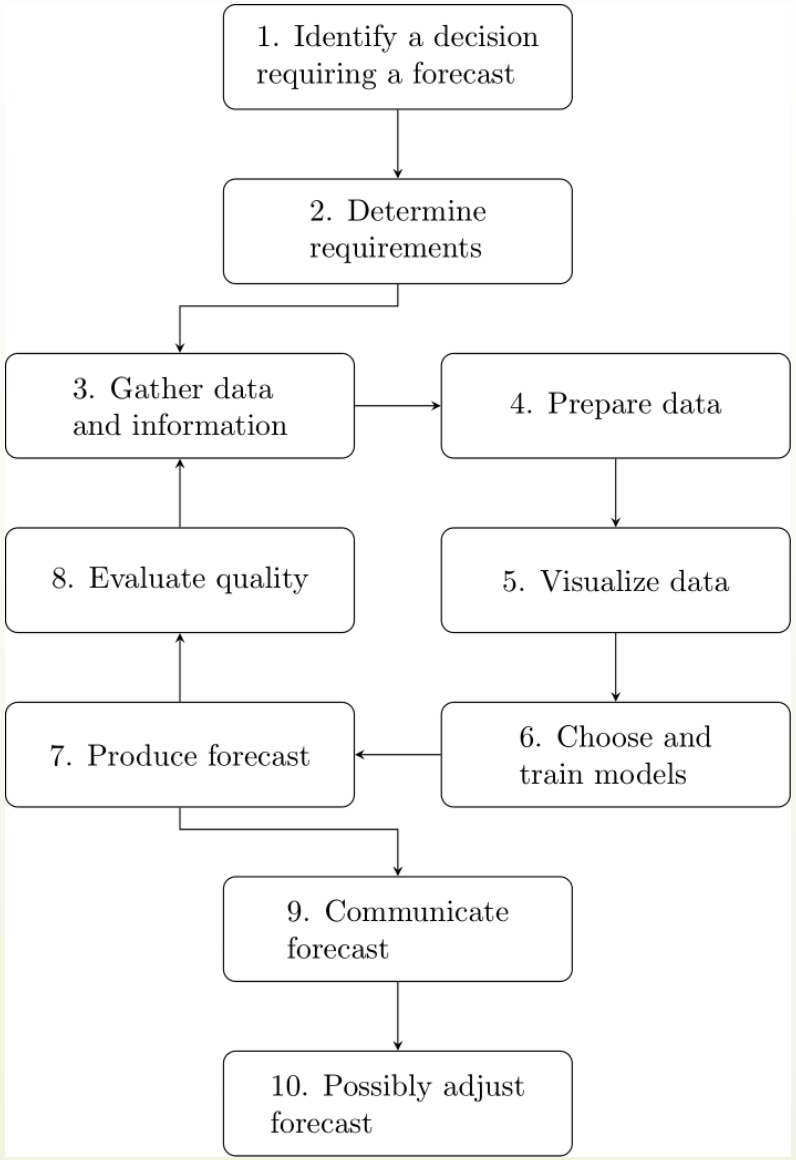
SECTION 1 OF 6

# Statistical forecasting

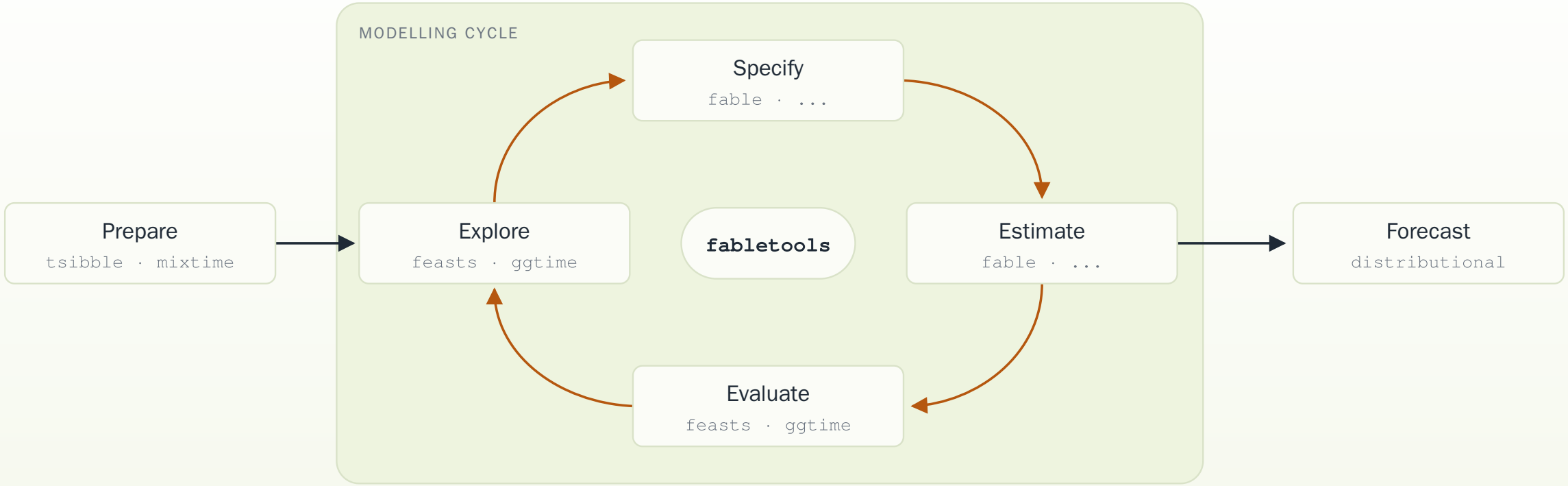
1 **Statistical forecasting** 2 What can we forecast? 3 Benchmark methods 4 Specify and estimate 5 Produce forecasts 6 Fitted values and residuals



# Forecasting workflow

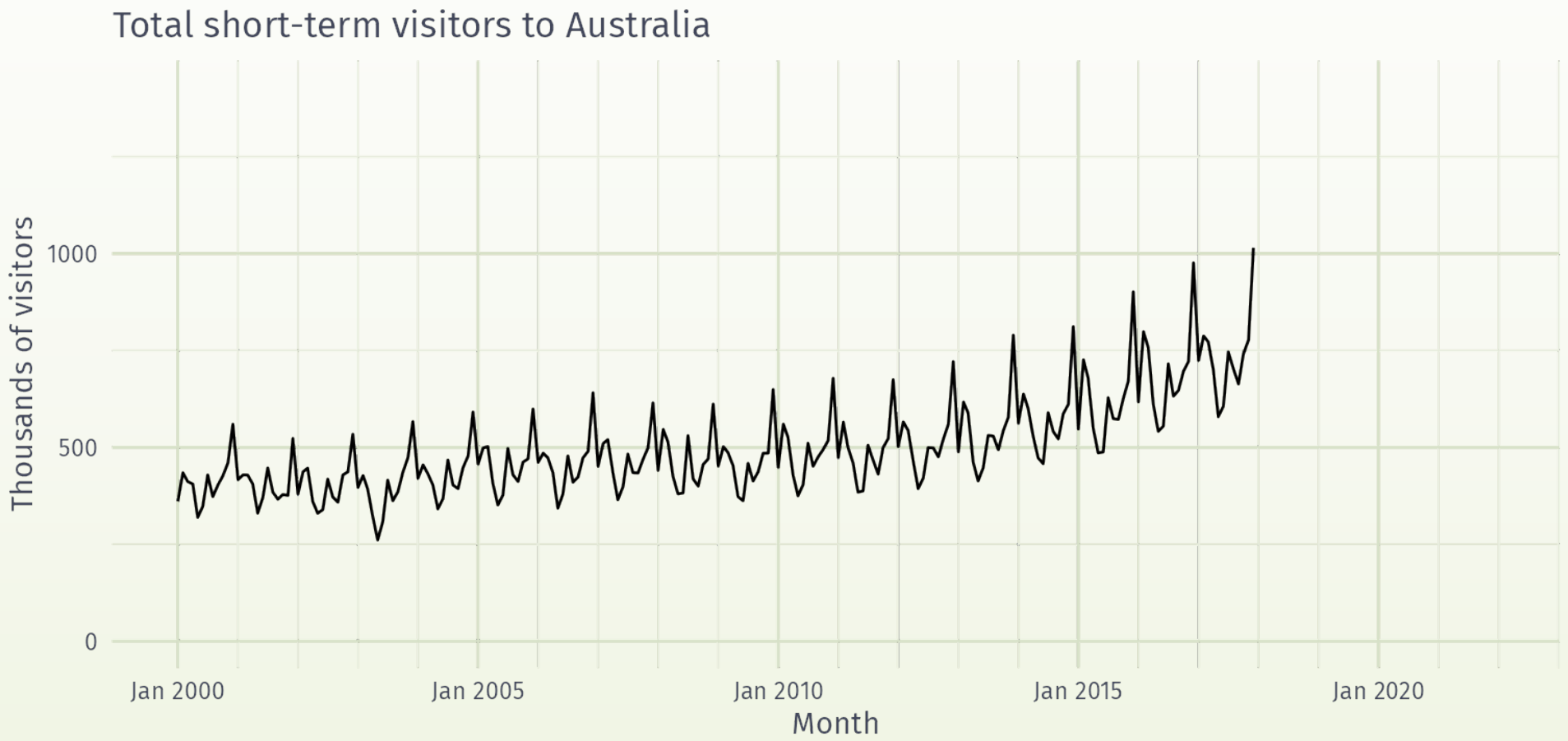


# Statistical forecasting steps



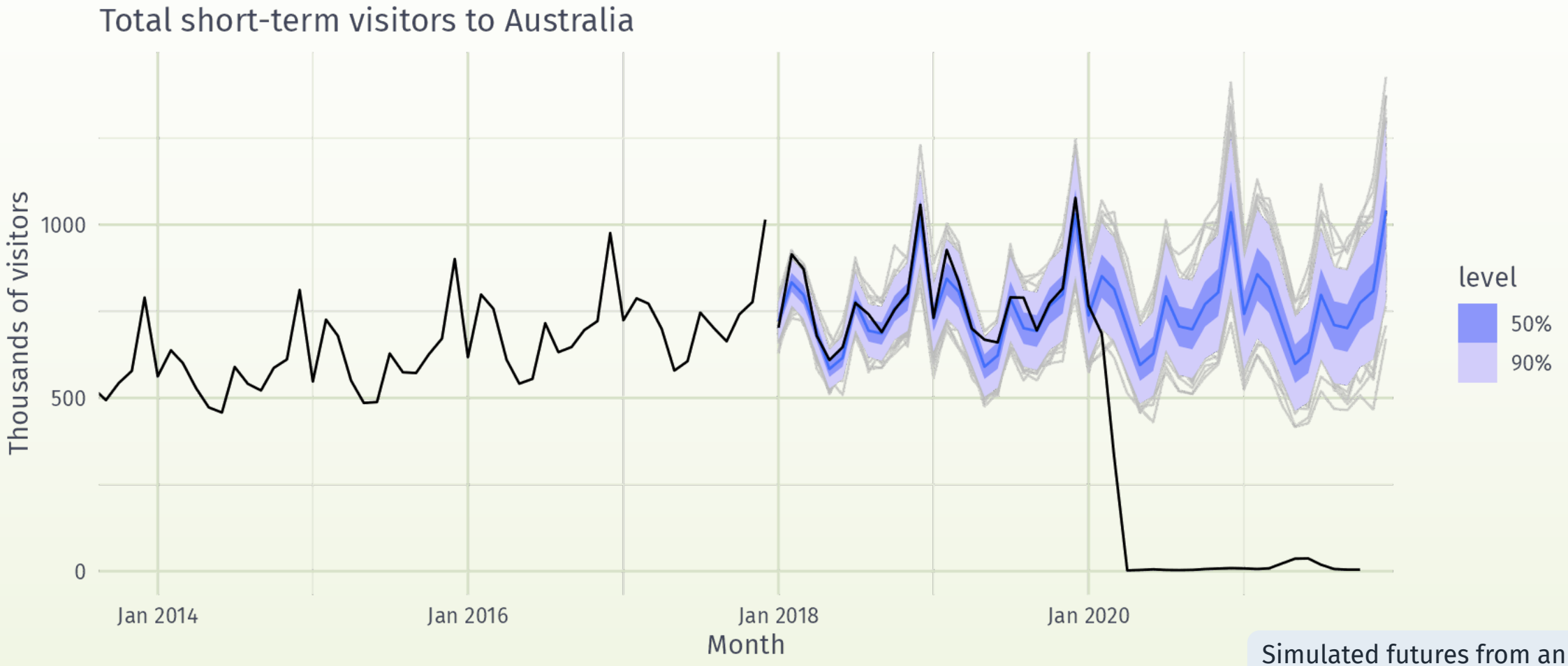
# What is a forecast?

A forecast is an estimate of the probability distribution of a variable to be observed in the future.



# What is a forecast?

A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# Was this a good forecast?



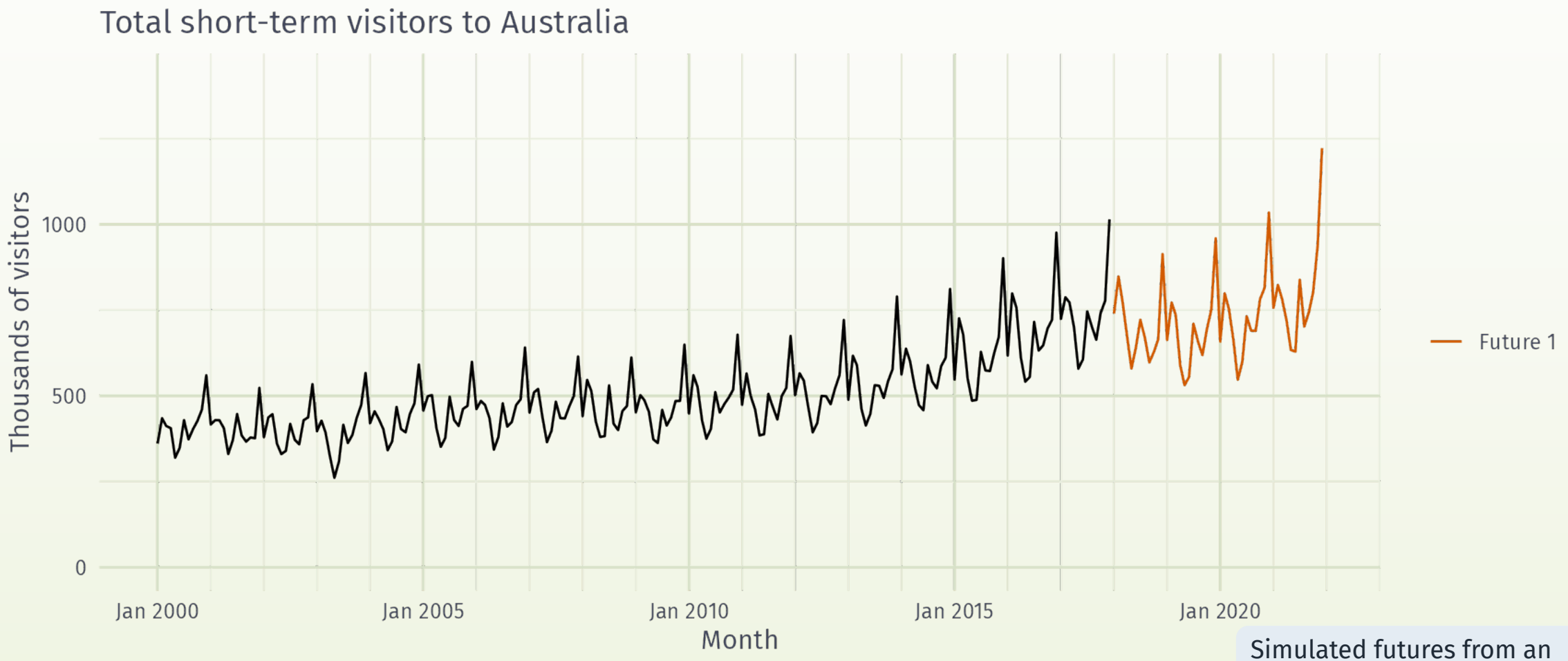

poll.nectric.com.au/s/
africast-modelling

**37** responses

● CLOSED 5 connected

# What is a forecast?

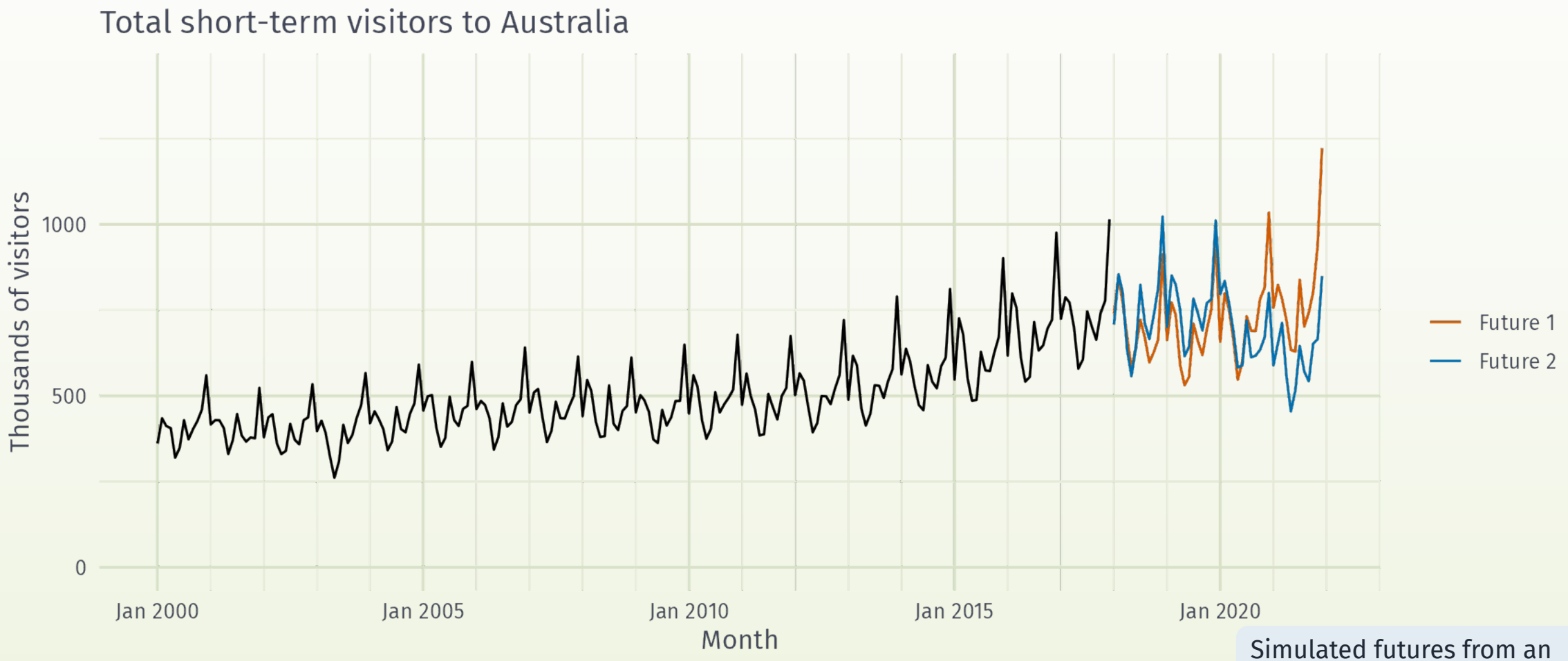
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# What is a forecast?

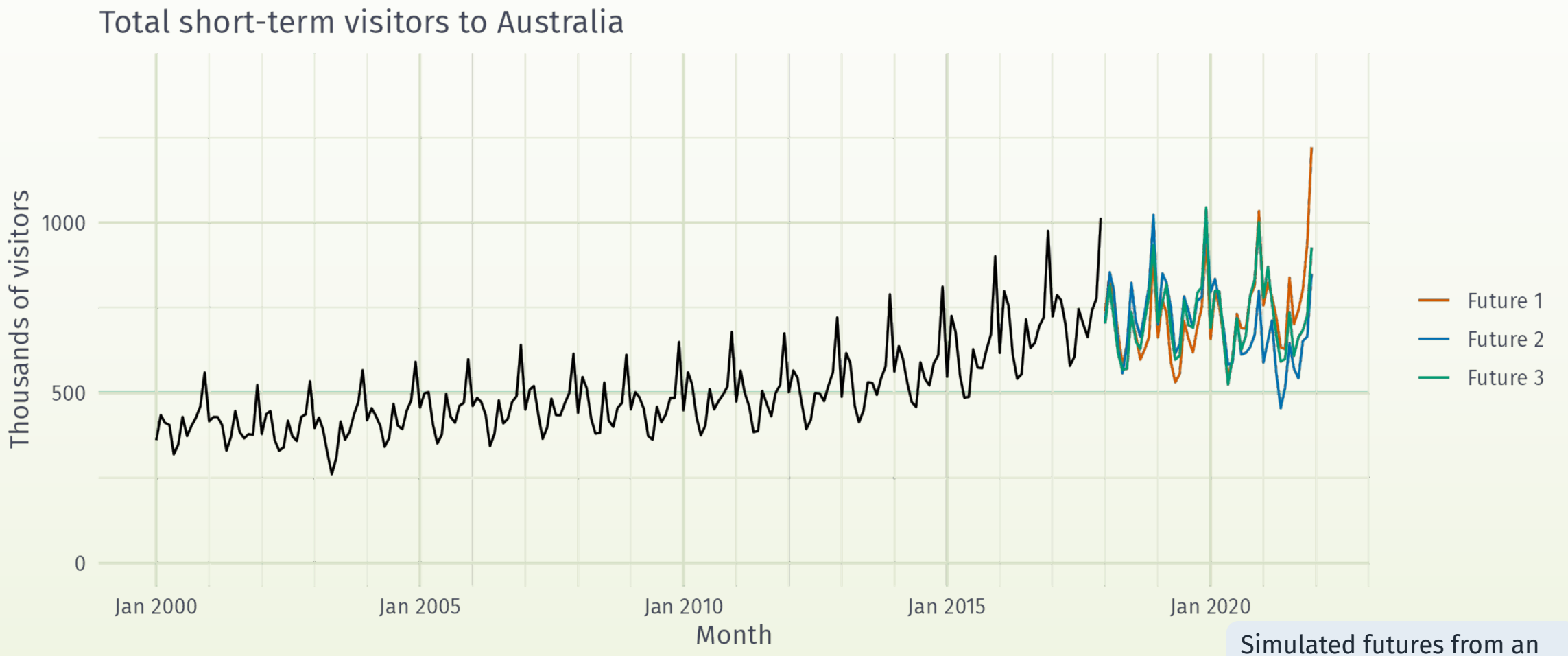
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# What is a forecast?

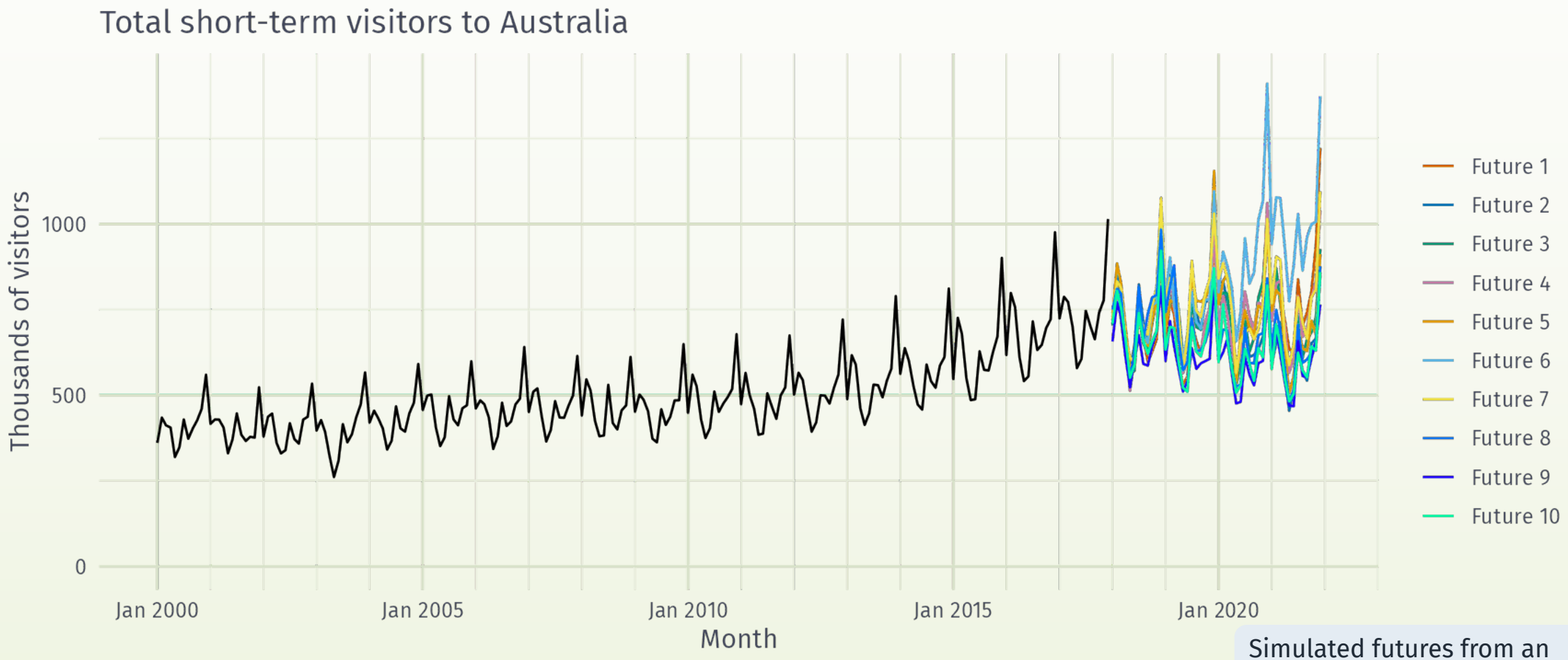
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# What is a forecast?

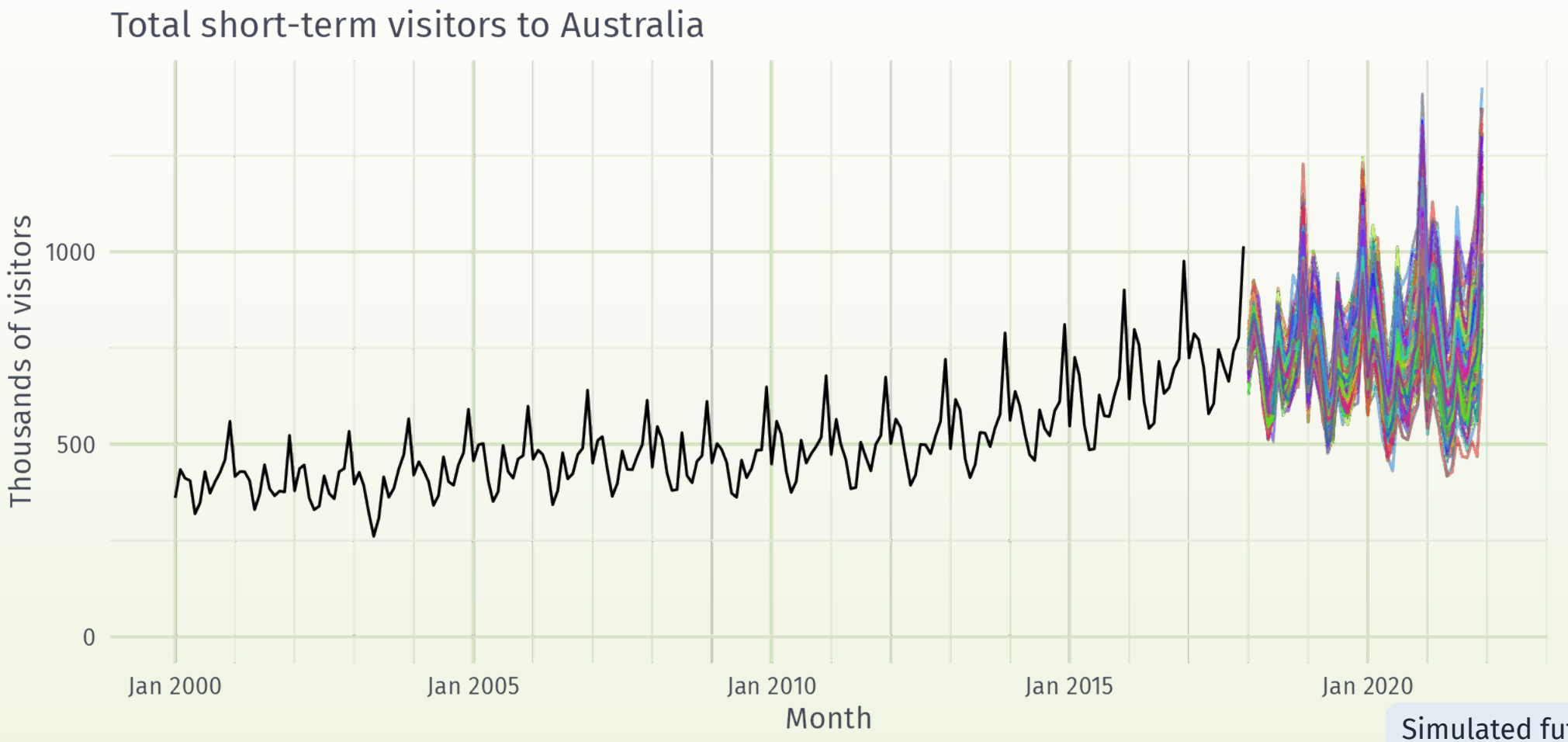
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# What is a forecast?

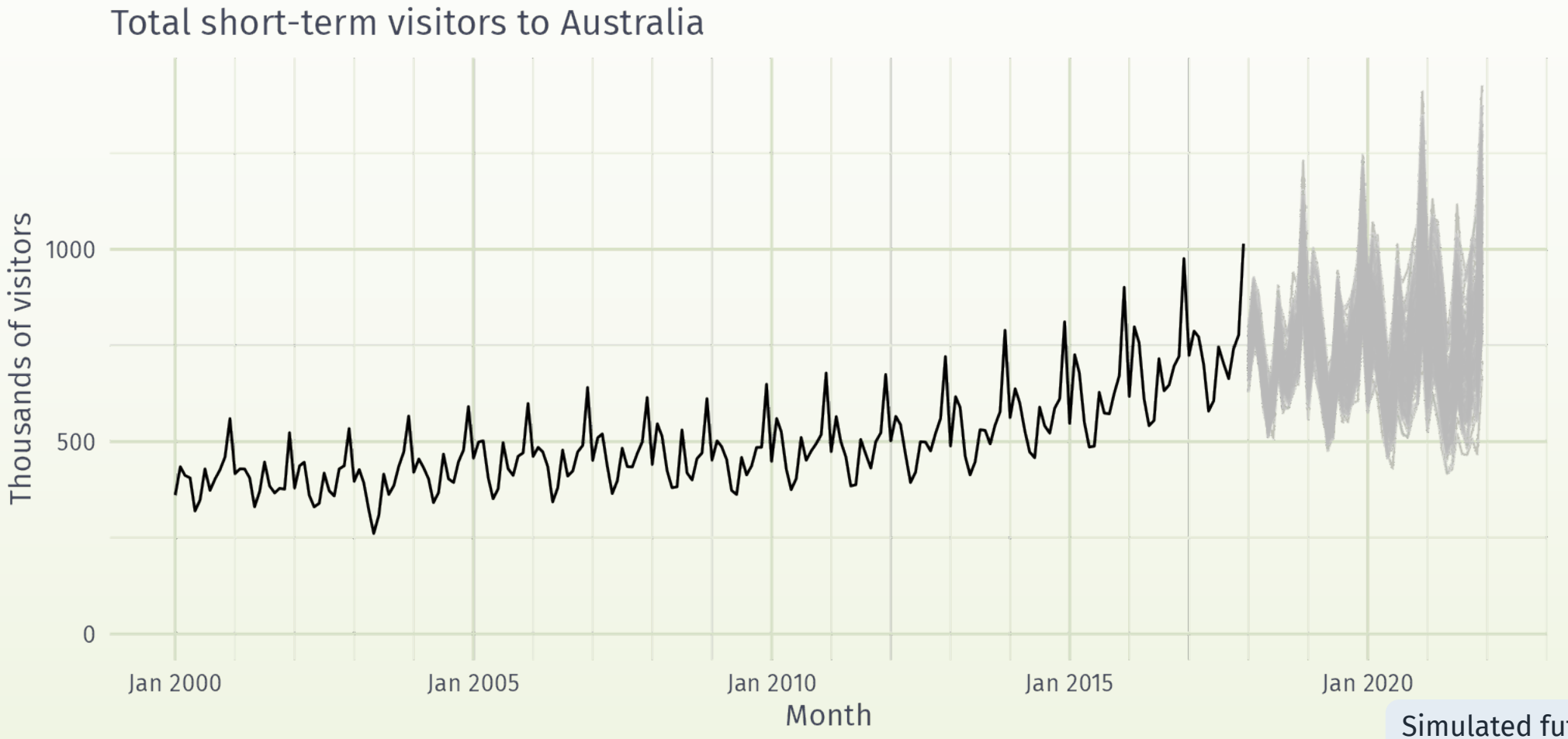
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# What is a forecast?

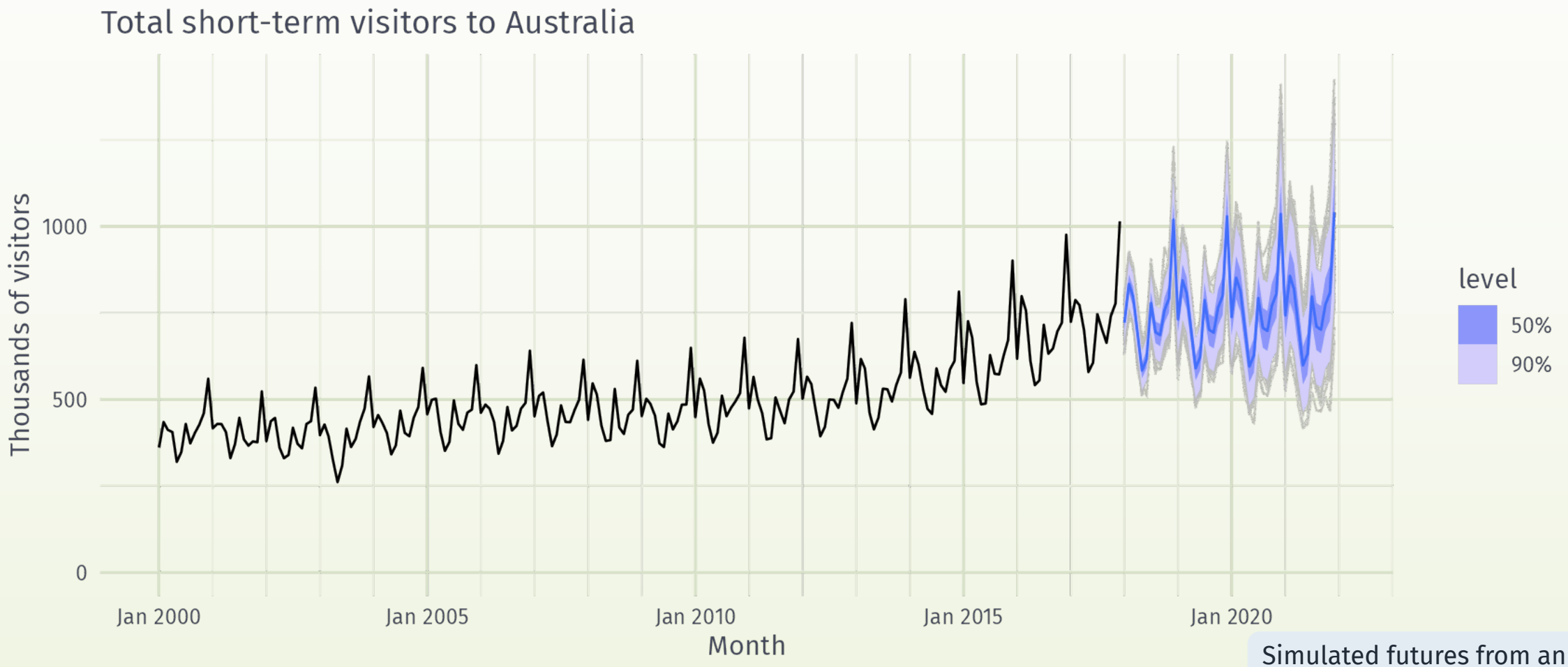
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

# What is a forecast?

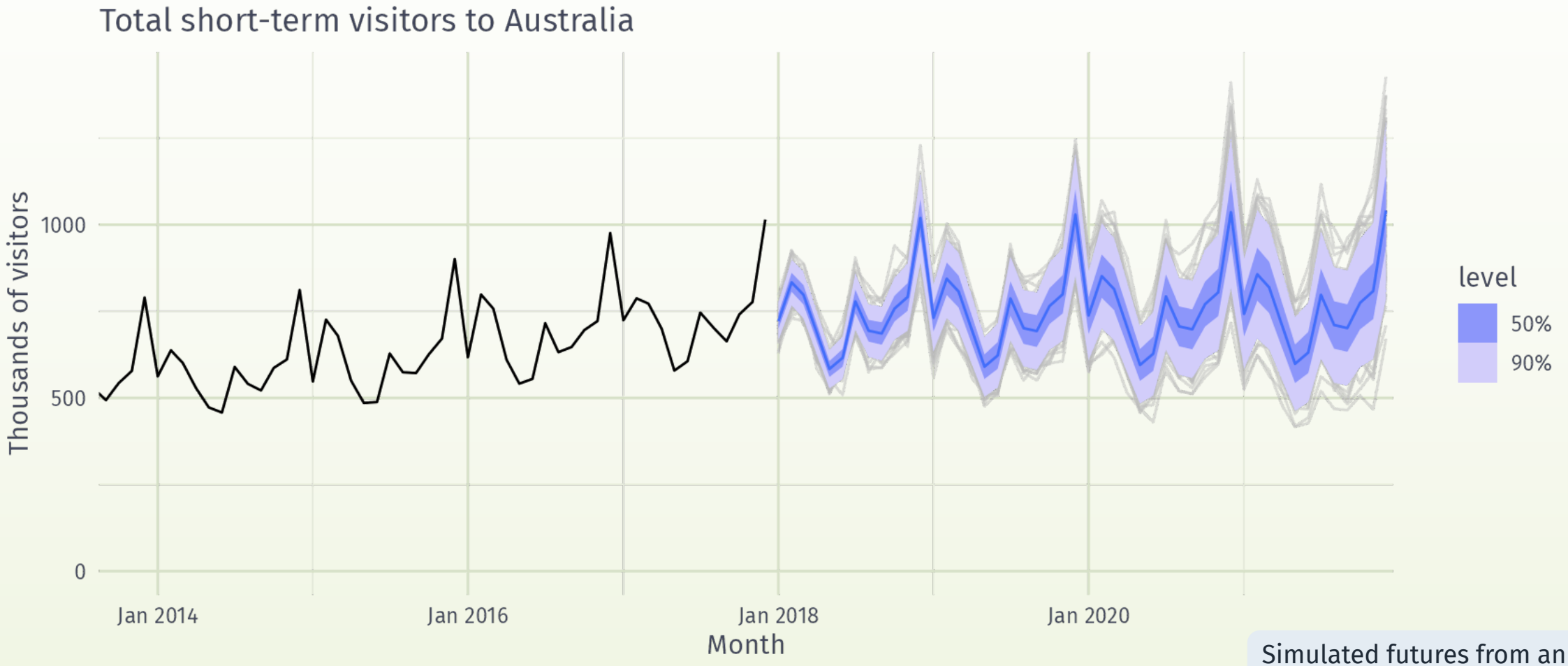
A forecast is an estimate of the probability distribution of a variable to be observed in the future.




Simulated futures from an ETS model

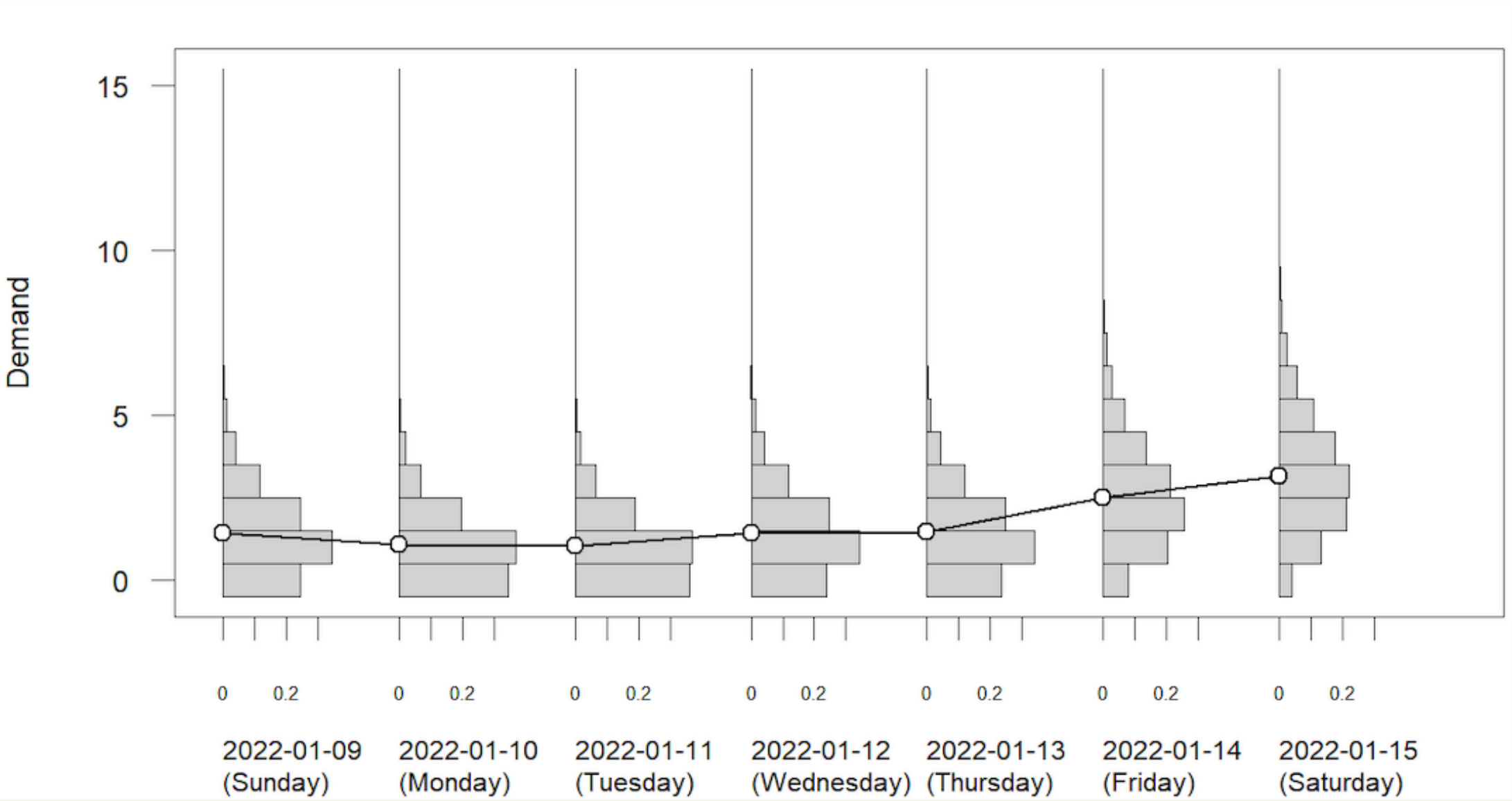
# Prediction interval

A 90% prediction interval contains the future value with 90% probability.

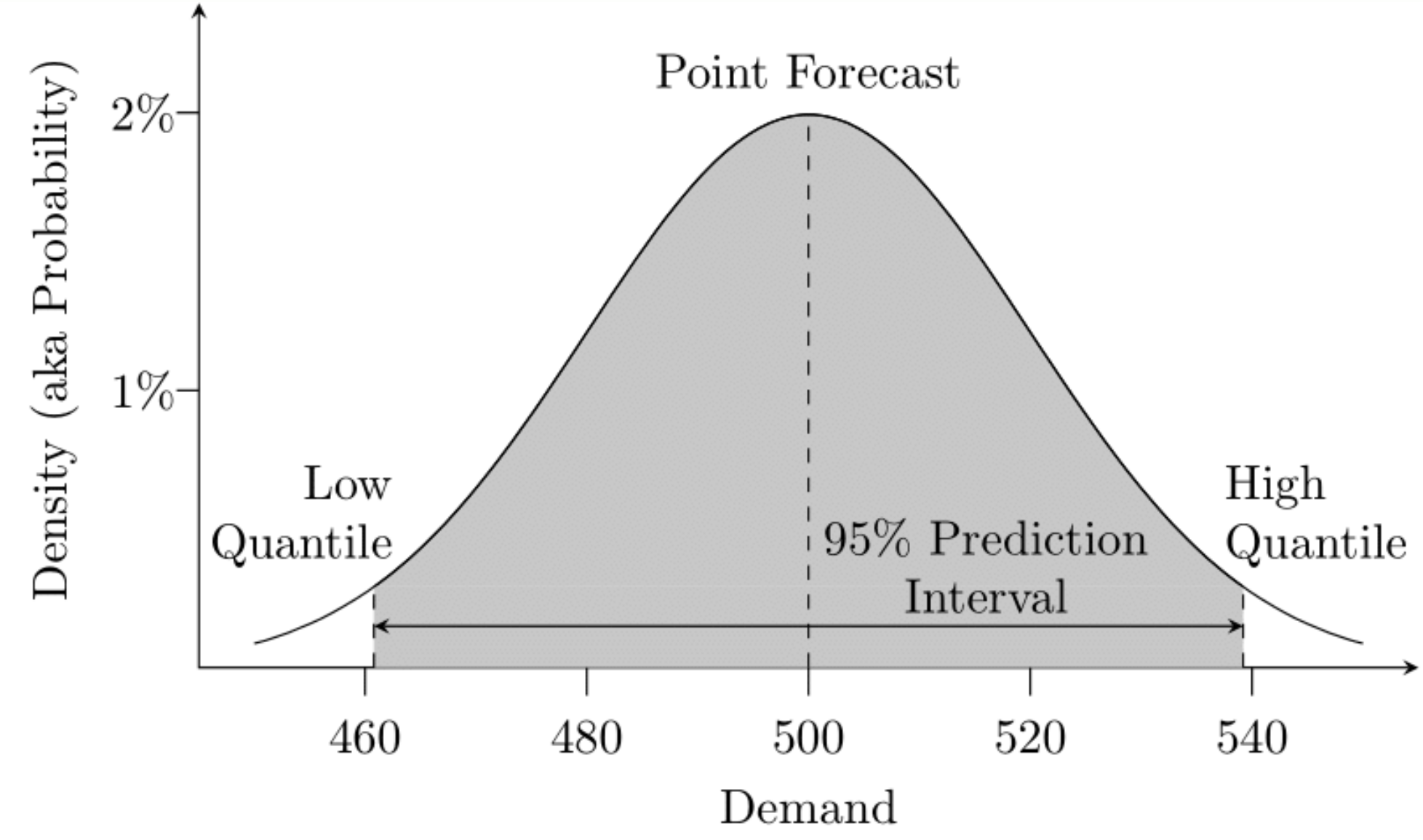



Simulated futures from an ETS model

# Visualising forecast distributions



# Forecast distribution



# Statistical forecasting

- Thing to be forecast: $y_{T+h}$.
- What we know: $y_1, \dots, y_T$.
- Forecast distribution: $y_{T+h|T} = y_{T+h} \mid \{y_1, y_2, \dots, y_T\}$.
- Point forecast: $\hat{y}_{T+h|T} = \mathrm{E}[y_{T+h} \mid y_1, \dots, y_T]$.
- Forecast variance: $\mathrm{Var}[y_{T+h} \mid y_1, \dots, y_T]$.
- Prediction interval is a range of values of $y_{T+h}$ with high probability.

SECTION 2 OF 6

# What can we forecast?

1 Statistical forecasting
2 **What can we forecast?**
3 Benchmark methods
4 Specify and estimate
5 Produce forecasts
6 Fitted values and residuals



# What can we forecast?



# What can we forecast?



# What can we forecast?



# What can we forecast?



# What can we forecast?



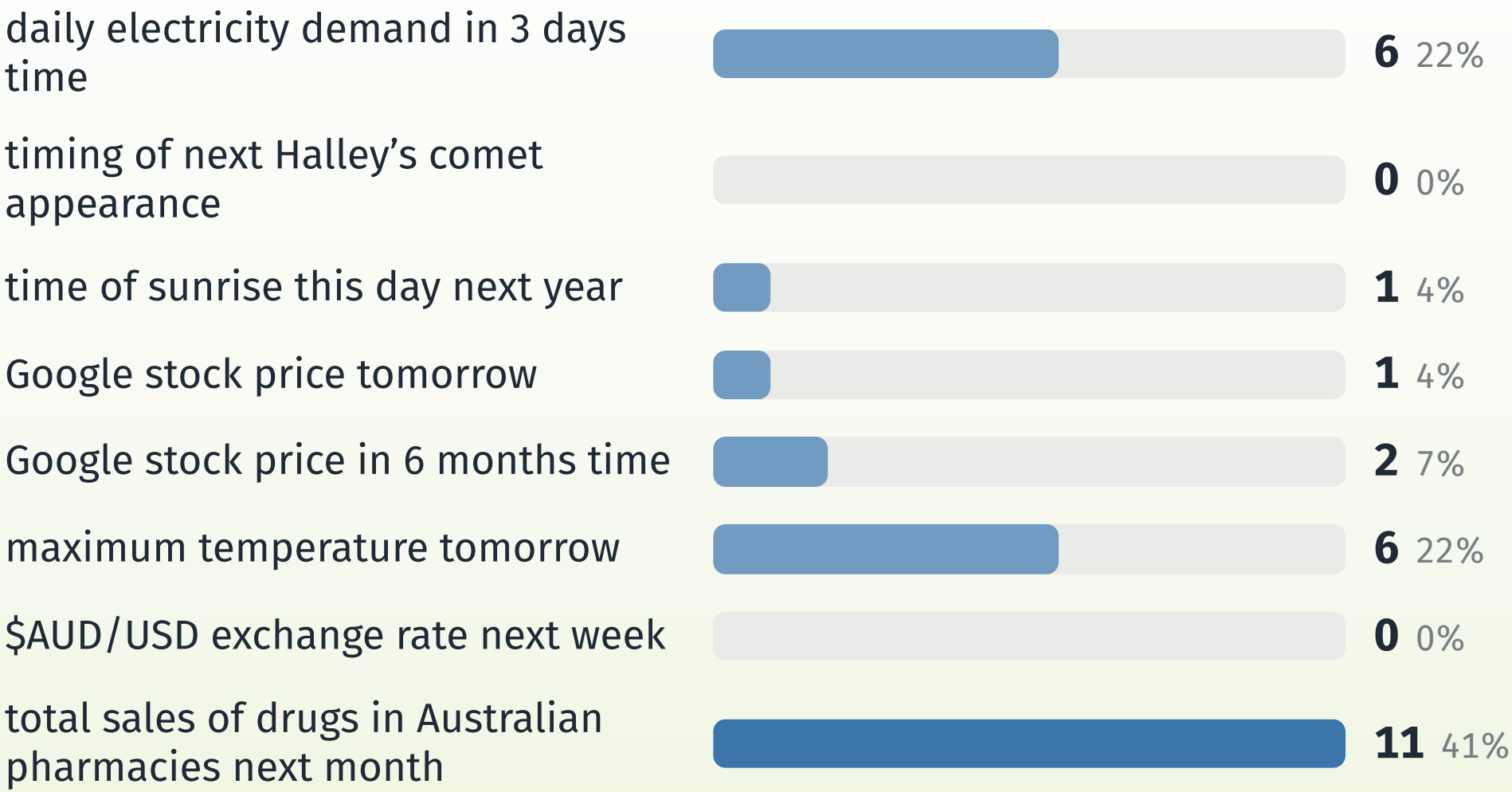
# What can we forecast?



# What can we forecast?



# Which is easiest to forecast?

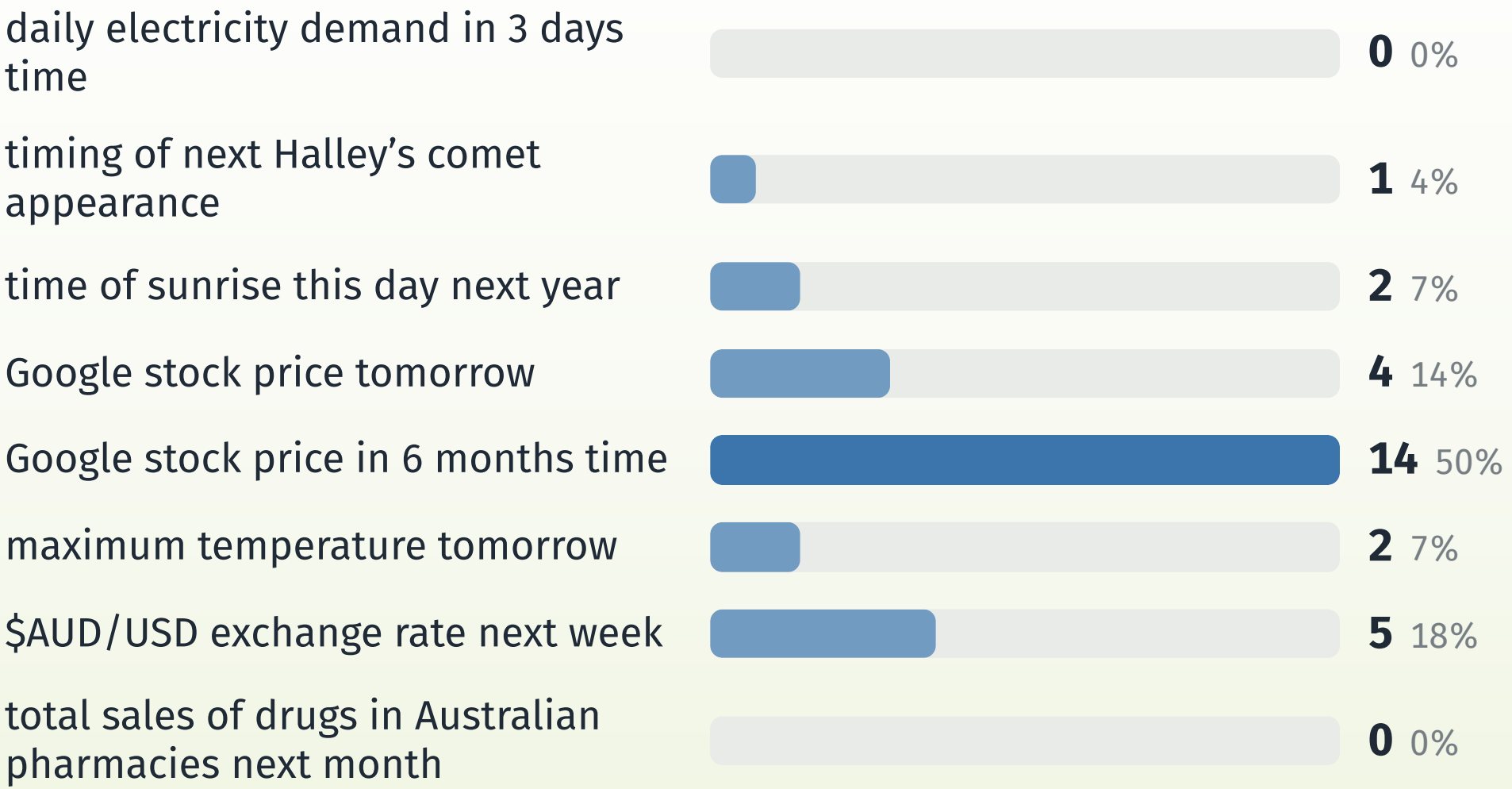



poll.nectric.com.au/s/africast-modelling

**27** responses

CLOSED 5 connected

# Which is hardest to forecast?




poll.nectric.com.au/s/africast-modelling

**28** responses

CLOSED 5 connected

# Factors affecting forecastability

Something is easier to forecast if:

- we have a good understanding of the factors that contribute to it;
- there is lots of data available;
- the forecasts cannot affect the thing we are trying to forecast;
- there is relatively low natural/unexplainable random variation;
- the future is somewhat similar to the past.

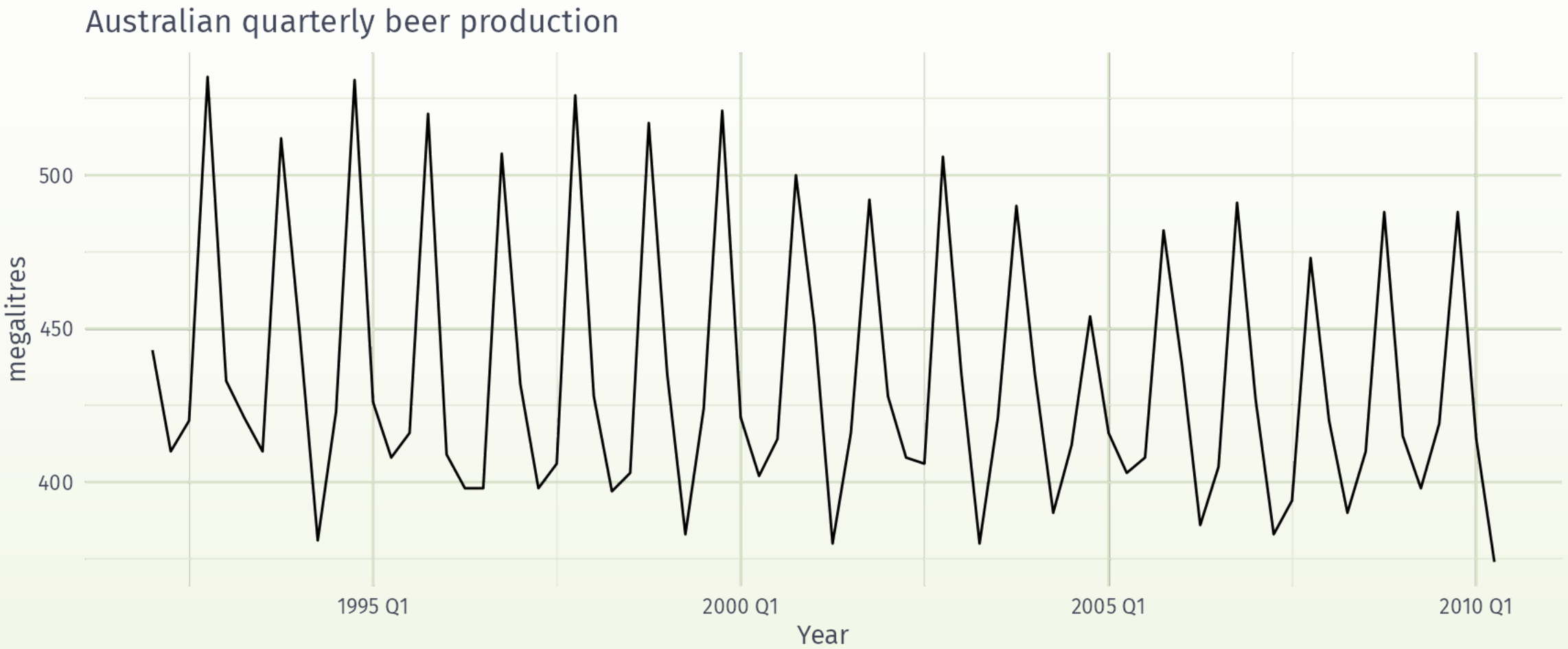
SECTION 3 OF 6

# Benchmark methods

1 Statistical forecasting
2 What can we forecast?
3 **Benchmark methods**
4 Specify and estimate
5 Produce forecasts
6 Fitted values and residuals

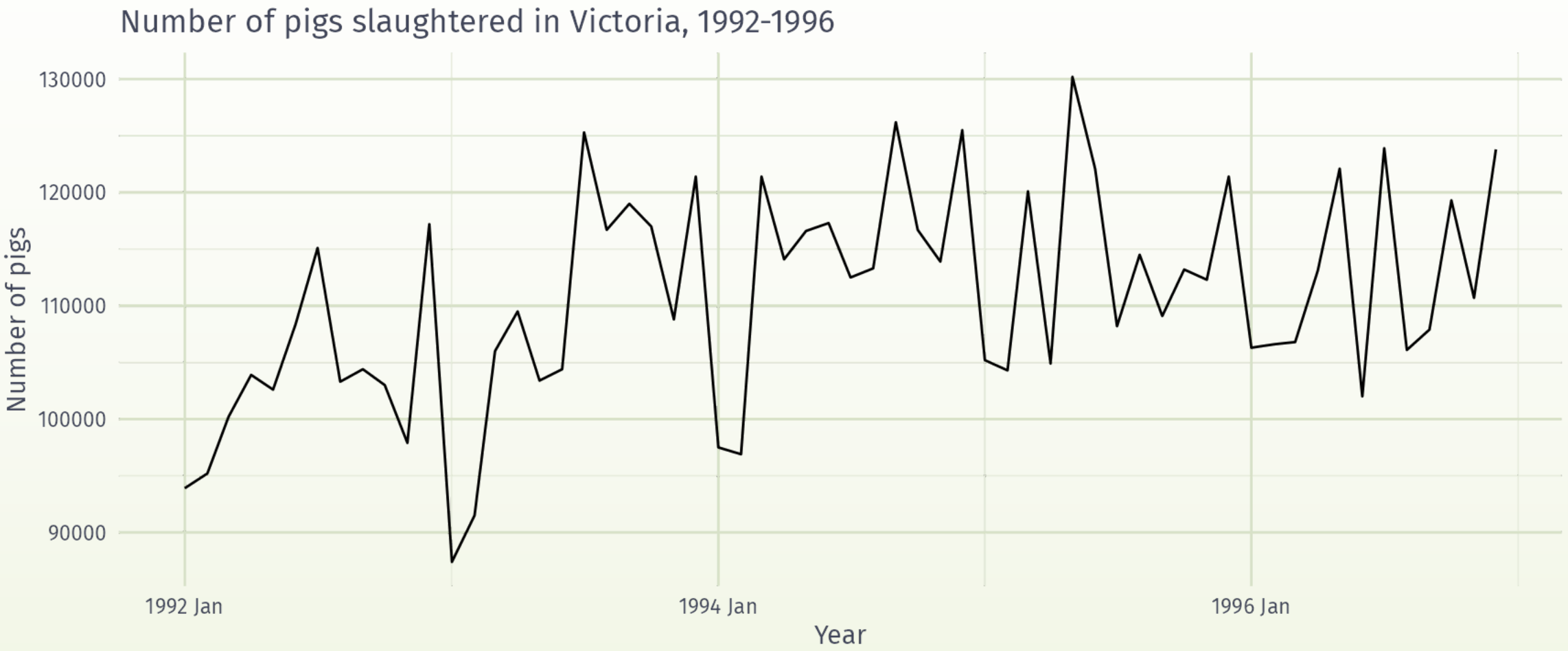


# Some simple forecasting methods




How would you forecast these series?

# Some simple forecasting methods




How would you forecast these series?

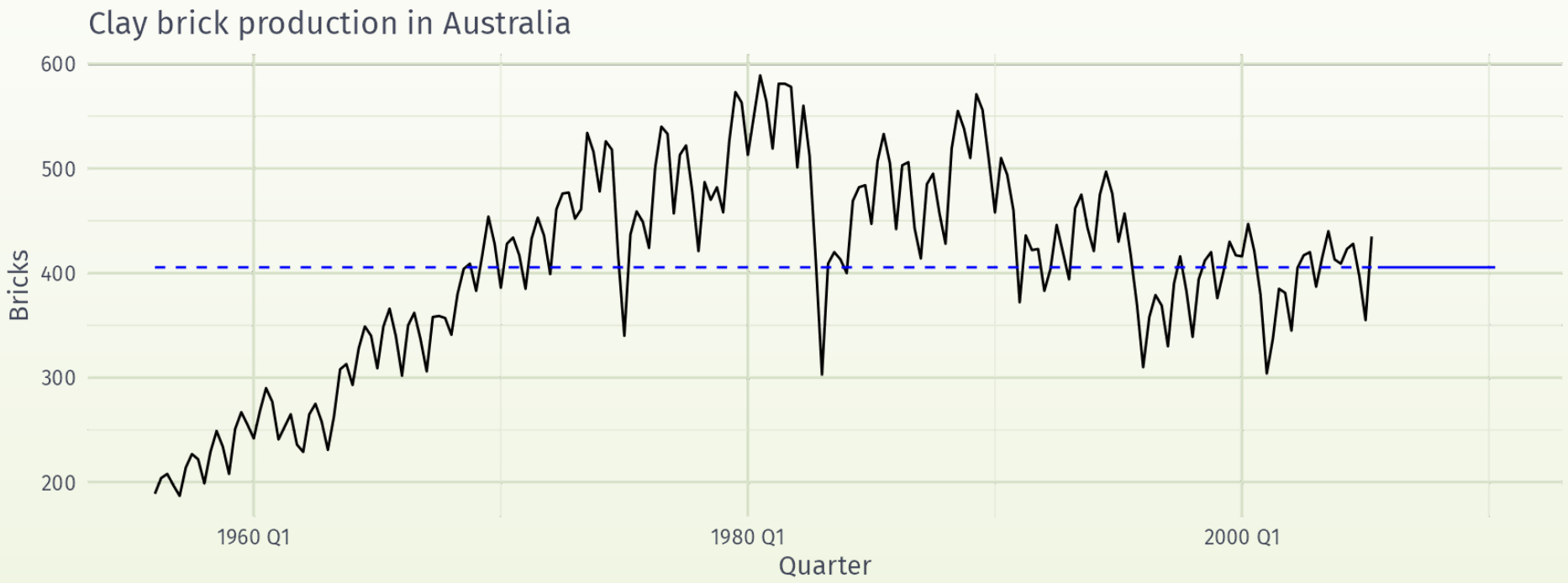
# Some simple forecasting methods




How would you forecast these series?

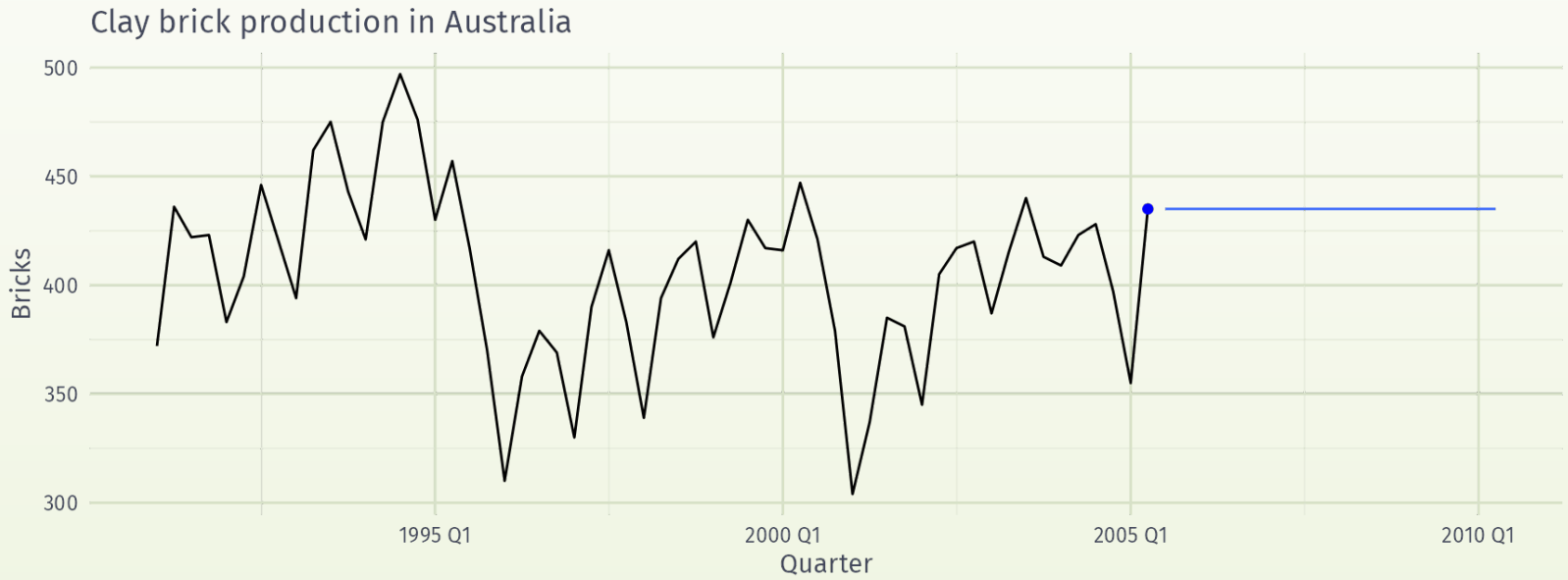
## `MEAN(y)`: Average method

- Forecast of all future values is equal to mean of historical data $\{y_1, \ldots, y_T\}$.
- Forecasts: $\hat{y}_{T+h|T} = \bar{y} = (y_1 + \cdots + y_T)/T$



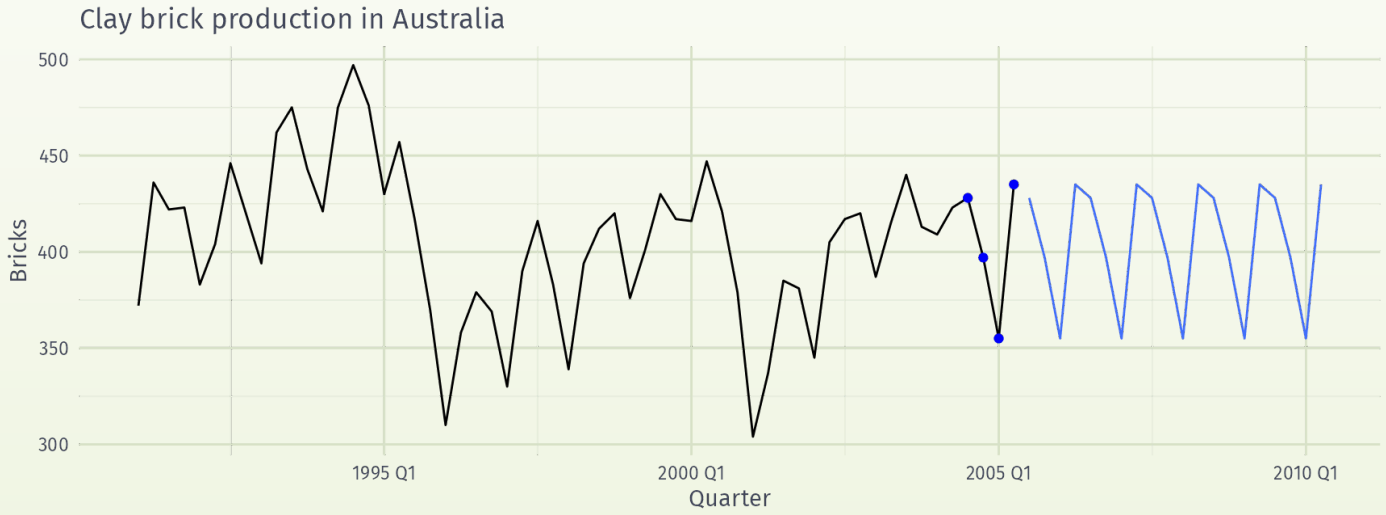
## `NAIVE(y)`: Naïve method

- Forecasts equal to last observed value.
- Forecasts: $\hat{y}_{T+h|T} = y_T$.
- Optimal when the series is a random walk (e.g. stock prices).



## `SNAIVE(y ~ lag(m))`: Seasonal naïve method

- Forecasts equal to last value from same season.
- Forecasts: $\hat{y}_{T+h|T} = y_{T+h-m(k+1)}$, where $m =$ seasonal period and $k$ is the integer part of $(h-1)/m$.
- `lag()` defaults to the seasonal period, so `SNAIVE(y)` is enough.



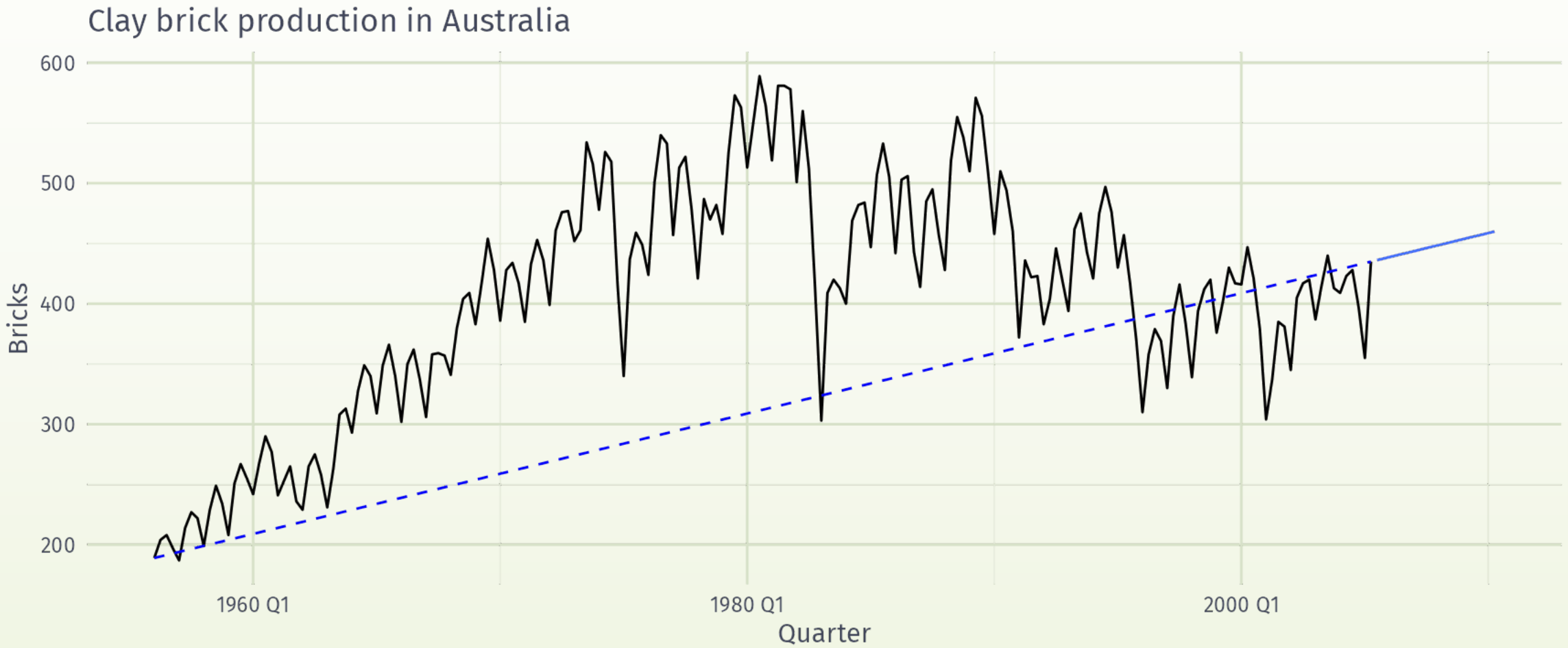
## `RW(y ~ drift())`: Drift method

- Forecasts equal to last value plus average change.
- Forecasts:

$$\begin{aligned}\hat{y}_{T+h|T} &= y_T + \frac{h}{T-1}\sum_{t=2}^{T}(y_t - y_{t-1}) \\ &= y_T + \frac{h}{T-1}(y_T - y_1).\end{aligned}$$

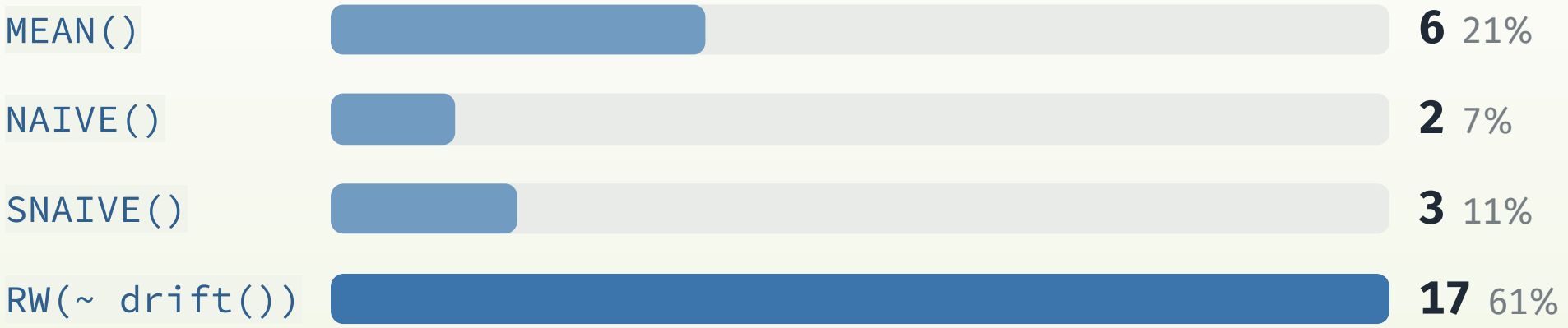
- Equivalent to extrapolating a line drawn between first and last observations.

# Drift method



# Which benchmark suits the pigs series?

Which benchmark method is best for the number of pigs slaughtered in Victoria?




poll.nectric.com.au/s/africast-modelling

**28** responses

● CLOSED 5 connected

SECTION 4 OF 6

# Specify and estimate

1 Statistical forecasting
2 What can we forecast?
3 Benchmark methods
4 **Specify and estimate**
5 Produce forecasts
6 Fitted values and residuals



# Model specification

- Model specification in fable supports a formula-based interface.
- A model formula in R is expressed using `response ~ terms`:
  - the left side describes the response;
  - the right side describes the terms used to model the response.
- Model names are in capitals, e.g. `SNAIVE(Beer ~ lag("year"))` or `RW(Beer ~ drift())`.

# Model fitting: example

The `model()` function trains models on data.

```
beer_fit <- aus_production |>
  model(
    Seasonal_naive = SNAIVE(Beer),
    Naive = NAIVE(Beer),
    Drift = RW(Beer ~ drift()),
    Mean = MEAN(Beer)
  )
```

```
# A mable: 1 x 4
  Seasonal_naive   Naive          Drift    Mean
          <model> <model>        <model> <model>
1       <SNAIVE> <NAIVE> <RW w/ drift>  <MEAN>
```

A `mable` is a model table, each cell corresponds to a fitted model.

# Extract information from `mable`

```
beer_fit |> select(Seasonal_naive) |> report()
beer_fit |> tidy()
beer_fit |> glance()
```

- `report()` gives a formatted model-specific display.
- `tidy()` extracts the estimated parameters (if any) of each model.
- `glance()` gives one row of summary statistics per model, such as `sigma2`.
- `select()` and `filter()` pick out specific models.

SECTION 5 OF 6

# Produce forecasts

1 Statistical forecasting
2 What can we forecast?
3 Benchmark methods
4 Specify and estimate
5 **Produce forecasts**
6 Fitted values and residuals



# Producing forecasts

- The `forecast()` function is used to produce forecasts from estimated models.
- The forecast horizon can be specified with:
  - `h` as a number (the number of future observations);
  - `h` as natural language (the length of time to predict, e.g. `"5 years"`);
  - `new_data`, a dataset of future time periods.

# Producing forecasts

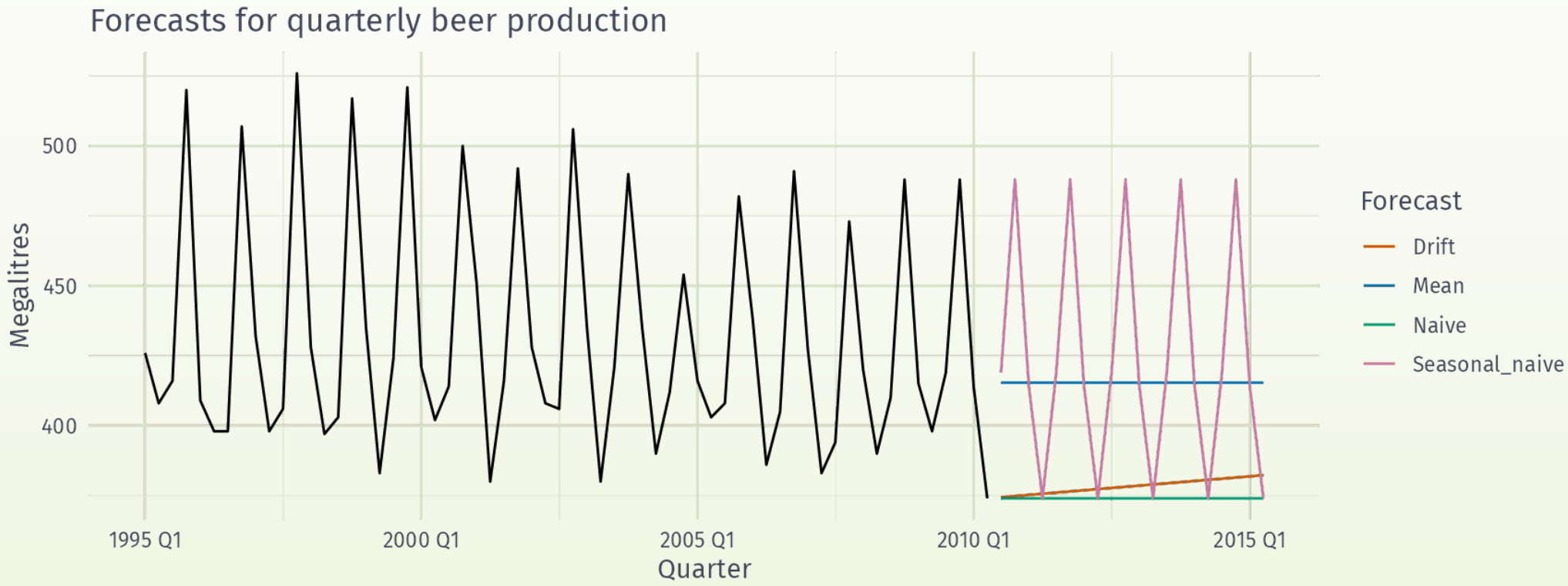
```
beer_fc <- beer_fit |>
  forecast(h = "5 years")
```

```
# A fable: 80 x 4 [1Q]
# Key:     .model [4]
  .model          Quarter         Beer .mean
  <chr>             <qtr>       <dist> <dbl>
1 Seasonal_naive 2010 Q3 N(419, 373)   419
2 Seasonal_naive 2010 Q4 N(488, 373)   488
3 Seasonal_naive 2011 Q1 N(414, 373)   414
4 Seasonal_naive 2011 Q2 N(374, 373)   374
# i 76 more rows
```

A `fable` is a forecast table with point forecasts and distributions.

# Visualising forecasts

```
beer_fc |>
  autoplot(filter(aus_production, year(Quarter) >= 1995), level = NULL) +
  labs(title = "Forecasts for quarterly beer production",
       x = "Quarter", y = "Megalitres") +
  guides(colour = guide_legend(title = "Forecast"))
```



# Forecast distributions

- A forecast $\hat{y}_{T+h|T}$ is (usually) the mean of the conditional distribution $y_{T+h} \mid y_1, \dots, y_T$.
- Many time series models produce normally distributed forecasts.
- The forecast distribution describes the probability of observing any future value.

# Forecast distributions - normal distribution

Assuming residuals are normal and uncorrelated, with variance $\hat{\sigma}^2$:

| | |
|---|---|
| **Mean:** | $y_{T+h\|T} \sim N(\bar{y}, (1+1/T)\hat{\sigma}^2)$ |
| **Naïve:** | $y_{T+h\|T} \sim N(y_T, h\hat{\sigma}^2)$ |
| **Seasonal naïve:** | $y_{T+h\|T} \sim N(y_{T+h-m(k+1)}, (k+1)\hat{\sigma}^2)$ |
| **Drift:** | $y_{T+h\|T} \sim N(y_T + \frac{h}{T-1}(y_T - y_1), h\frac{T+h}{T}\hat{\sigma}^2)$ |

where $k$ is the integer part of $(h-1)/m$.

Note that when $h=1$ and $T$ is large, these all give the same approximate forecast variance: $\hat{\sigma}^2$.

# Forecast distributions from bootstrapping

When a normal distribution for the residuals is an unreasonable assumption, one alternative is to use bootstrapping, which only assumes that the residuals are uncorrelated with constant variance.

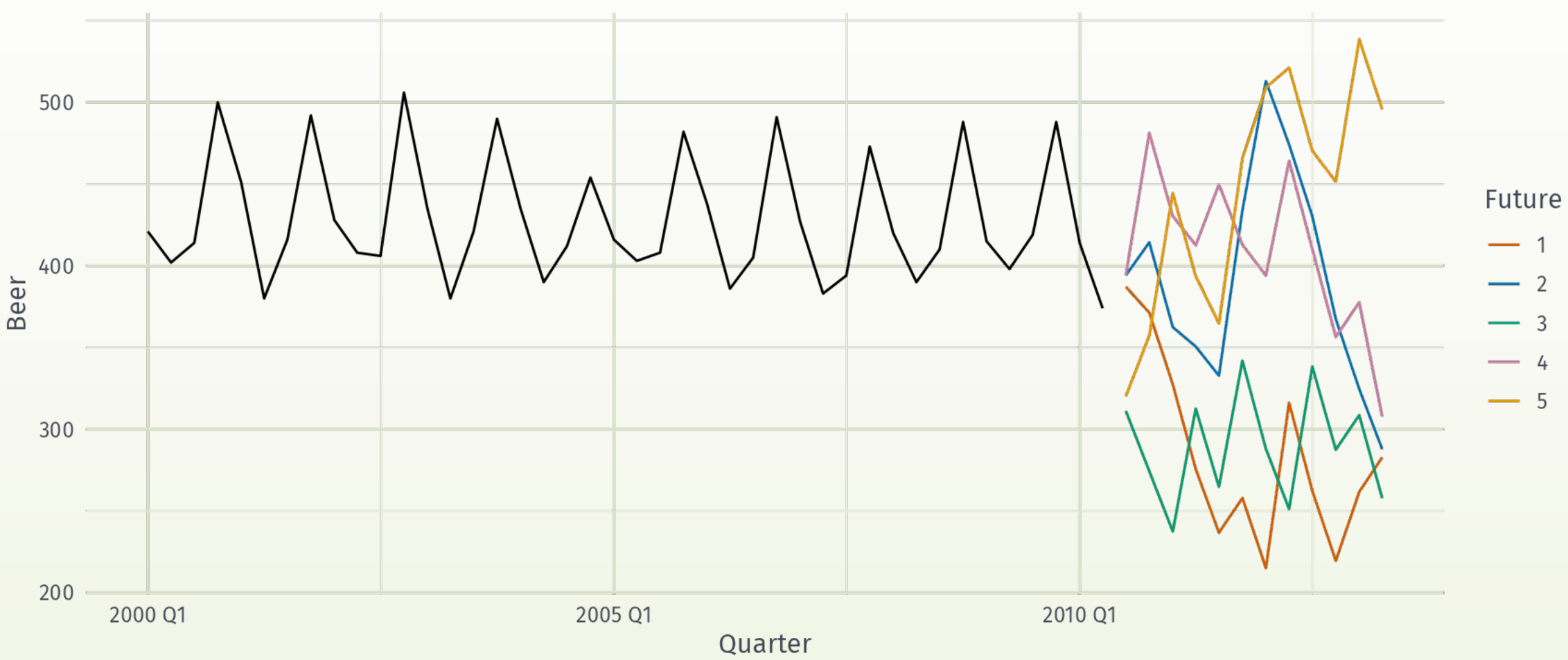
- A residual (one-step forecast error) is $e_t = y_t - \hat{y}_{t|t-1}$, so $y_t = \hat{y}_{t|t-1} + e_t$.
- So we can simulate the next observation of a time series using $y_{T+1} = \hat{y}_{T+1|T} + e_{T+1}$, where $e_{T+1}$ is sampled (with replacement) from the past residuals.
- Adding the new simulated observation to our data set, we can repeat the process to obtain $y_{T+2} = \hat{y}_{T+2|T+1} + e_{T+2}$

# Generate many possible futures using `generate()`

```
beer_2000 <- aus_production |> filter(year(Quarter) >= 2000) |> select(Beer)
beer_2000_fit <- beer_2000 |>
  model(NAIVE(Beer))
set.seed(2024)
beer_2000_sim <- beer_2000_fit |> generate(h = 12, times = 5, bootstrap = TRUE)
print(beer_2000_sim, n = 5)
```

```
# A tsibble: 60 x 4 [1Q]
# Key:       .model, .rep [5]
  .model      Quarter .rep   .sim
  <chr>         <qtr> <chr> <dbl>
1 NAIVE(Beer) 2010 Q3 1      387.
2 NAIVE(Beer) 2010 Q4 1      371.
3 NAIVE(Beer) 2011 Q1 1      327.
4 NAIVE(Beer) 2011 Q2 1      276.
5 NAIVE(Beer) 2011 Q3 1      237.
# i 55 more rows
```

# Generate 5 different futures



# Use `forecast()` to produce probabilistic forecasts

```
beer_2000_fc <- beer_2000_fit |> forecast(h = 12, bootstrap = TRUE)
beer_2000_fc
```

```
# A fable: 12 x 4 [1Q]
# Key:     .model [1]
   .model       Quarter          Beer .mean
   <chr>          <qtr>        <dist> <dbl>
 1 NAIVE(Beer) 2010 Q3 sample[5000]  374.
 2 NAIVE(Beer) 2010 Q4 sample[5000]  375.
 3 NAIVE(Beer) 2011 Q1 sample[5000]  375.
 4 NAIVE(Beer) 2011 Q2 sample[5000]  374.
 5 NAIVE(Beer) 2011 Q3 sample[5000]  374.
 6 NAIVE(Beer) 2011 Q4 sample[5000]  374.
 7 NAIVE(Beer) 2012 Q1 sample[5000]  375.
 8 NAIVE(Beer) 2012 Q2 sample[5000]  374.
 9 NAIVE(Beer) 2012 Q3 sample[5000]  376.
10 NAIVE(Beer) 2012 Q4 sample[5000]  376.
11 NAIVE(Beer) 2013 Q1 sample[5000]  375.
12 NAIVE(Beer) 2013 Q2 sample[5000]  375.
```

# Prediction intervals

- Forecast intervals can be extracted using the `hilo()` function.

```
beer_naive_fit <- aus_production |> select(Beer) |> model(NAIVE(Beer))
forecast(beer_naive_fit, h = 8) |>
  hilo(level = 95) |>
  select(Quarter, .mean, `95%`)
```

```
# A tsibble: 8 x 3 [1Q]
  Quarter .mean           `95%`
    <qtr> <dbl>          <hilo>
1 2010 Q3   374 [241.36, 507]95
2 2010 Q4   374 [186.42, 562]95
3 2011 Q1   374 [144.27, 604]95
4 2011 Q2   374 [108.73, 639]95
5 2011 Q3   374 [ 77.42, 671]95
6 2011 Q4   374 [ 49.11, 699]95
7 2012 Q1   374 [ 23.08, 725]95
8 2012 Q2   374 [ -1.15, 749]95
```

# Prediction intervals

```
beer_fc |>
  filter(.model == "Seasonal_naive") |>
  hilo(level = c(50, 75)) |>
  select(Quarter, .mean, `50%`, `75%`)
```

```
# A tsibble: 20 x 4 [1Q]
   Quarter .mean         `50%`         `75%`
     <qtr> <dbl>       <hilo>       <hilo>
 1 2010 Q3   419 [406, 432]50 [397, 441]75
 2 2010 Q4   488 [475, 501]50 [466, 510]75
 3 2011 Q1   414 [401, 427]50 [392, 436]75
 4 2011 Q2   374 [361, 387]50 [352, 396]75
 5 2011 Q3   419 [401, 437]50 [388, 450]75
 6 2011 Q4   488 [470, 506]50 [457, 519]75
 7 2012 Q1   414 [396, 432]50 [383, 445]75
 8 2012 Q2   374 [356, 392]50 [343, 405]75
 9 2012 Q3   419 [396, 442]50 [380, 458]75
10 2012 Q4   488 [465, 511]50 [449, 527]75
11 2013 Q1   414 [391, 437]50 [375, 453]75
12 2013 Q2   374 [351, 397]50 [335, 413]75
13 2013 Q3   419 [393, 445]50 [375, 463]75
14 2013 Q4   488 [462, 514]50 [444, 532]75
15 2014 Q1   414 [388, 440]50 [370, 458]75
16 2014 Q2   374 [348, 400]50 [330, 418]75
17 2014 Q3   419 [390, 448]50 [369, 469]75
18 2014 Q4   488 [459, 517]50 [438, 538]75
```

# Prediction intervals

```
beer_fc |>
  filter(.model == "Seasonal_naive") |>
  hilo(level = 50) |>
  unpack_hilo(`50%`) |>
  select(Quarter, `50%_lower`, `50%_upper`)
```

```
# A tsibble: 20 x 3 [1Q]
   Quarter `50%_lower` `50%_upper`
     <qtr>       <dbl>       <dbl>
 1 2010 Q3        406.        432.
 2 2010 Q4        475.        501.
 3 2011 Q1        401.        427.
 4 2011 Q2        361.        387.
 5 2011 Q3        401.        437.
 6 2011 Q4        470.        506.
 7 2012 Q1        396.        432.
 8 2012 Q2        356.        392.
 9 2012 Q3        396.        442.
10 2012 Q4        465.        511.
11 2013 Q1        391.        437.
12 2013 Q2        351.        397.
13 2013 Q3        393.        445.
14 2013 Q4        462.        514.
15 2014 Q1        388.        440.
16 2014 Q2        348.        400.
17 2014 Q3        390.        448.
```

SECTION 6 OF 6

# Fitted values and residuals


1 Statistical forecasting
2 What can we forecast?
3 Benchmark methods
4 Specify and estimate
5 Produce forecasts
6 **Fitted values and residuals**




# Fitted values

- $\hat{y}_{t|t-1}$ is the forecast of $y_t$ based on observations $y_1, \ldots, y_{t-1}$.
- We call these “fitted values”.
- Sometimes drop the subscript: $\hat{y}_t \equiv \hat{y}_{t|t-1}$.
- Often not true forecasts since parameters are estimated on all data.

## For example:

- $\hat{y}_t = \bar{y}$ for average method.
- $\hat{y}_t = y_{t-1}$ for naïve method.
- $\hat{y}_t = y_{t-m}$ for seasonal naïve method.
- $\hat{y}_t = y_{t-1} + (y_T - y_1)/(T-1)$ for drift method.

# Forecasting residuals

**Residuals in forecasting:** difference between observed value and its fitted value: $e_t = y_t - \hat{y}_{t|t-1}$.

- Residuals are one-step errors on the training data; forecast errors are on unseen test data.
- `augment()` gives both `.resid` and `.innov` (innovation residuals); they are equal unless the response is transformed.

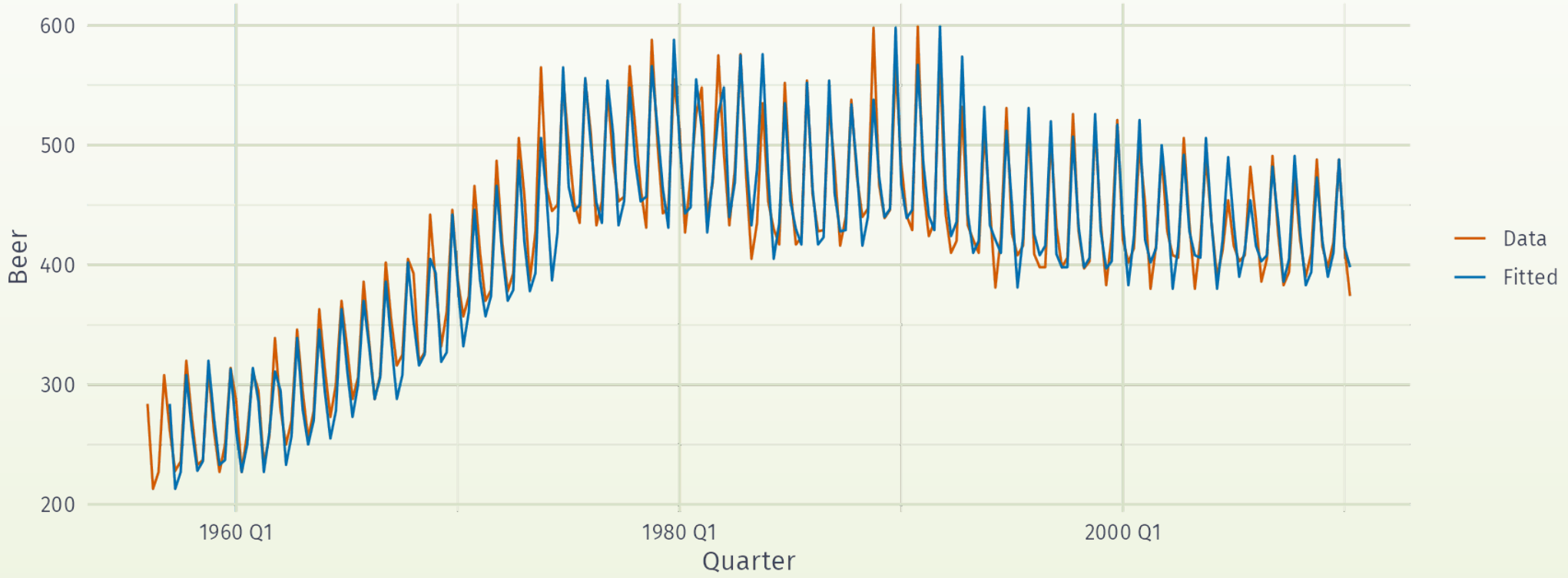
# Beer production - augment

```
beer_snaive_fit <- aus_production |> select(Beer) |> model(SNAIVE(Beer))
augment(beer_snaive_fit)
```

```
# A tsibble: 218 x 6 [1Q]
# Key:       .model [1]
   .model        Quarter  Beer .fitted .resid .innov
   <chr>           <qtr> <dbl>   <dbl>  <dbl>  <dbl>
 1 SNAIVE(Beer) 1956 Q1   284      NA     NA     NA
 2 SNAIVE(Beer) 1956 Q2   213      NA     NA     NA
 3 SNAIVE(Beer) 1956 Q3   227      NA     NA     NA
 4 SNAIVE(Beer) 1956 Q4   308      NA     NA     NA
 5 SNAIVE(Beer) 1957 Q1   262     284    -22    -22
 6 SNAIVE(Beer) 1957 Q2   228     213     15     15
 7 SNAIVE(Beer) 1957 Q3   236     227      9      9
 8 SNAIVE(Beer) 1957 Q4   320     308     12     12
 9 SNAIVE(Beer) 1958 Q1   272     262     10     10
10 SNAIVE(Beer) 1958 Q2   233     228      5      5
# i 208 more rows
```

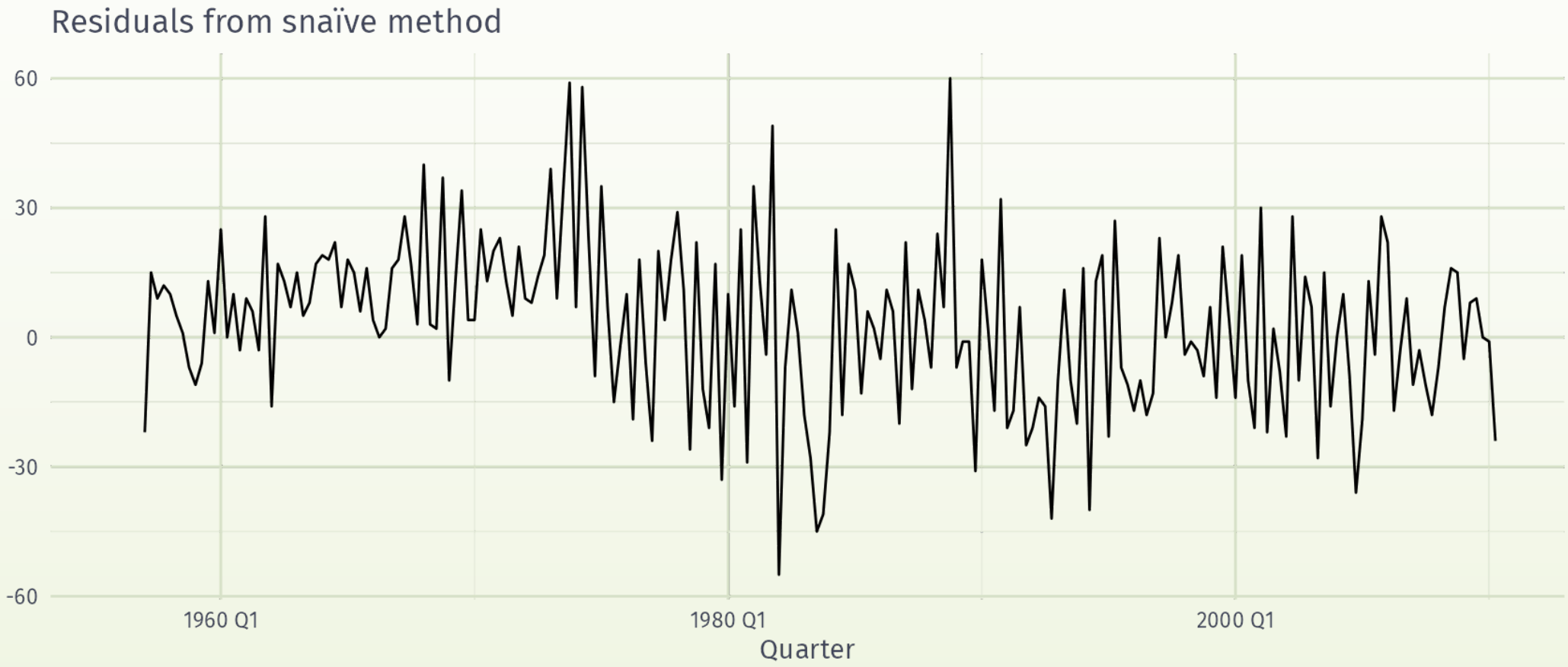
# Beer production - fitted values

```
augment(beer_snaive_fit) |>
  ggplot(aes(x = Quarter)) +
  geom_line(aes(y = Beer, colour = "Data")) +
  geom_line(aes(y = .fitted, colour = "Fitted")) +
  labs(colour = NULL)
```



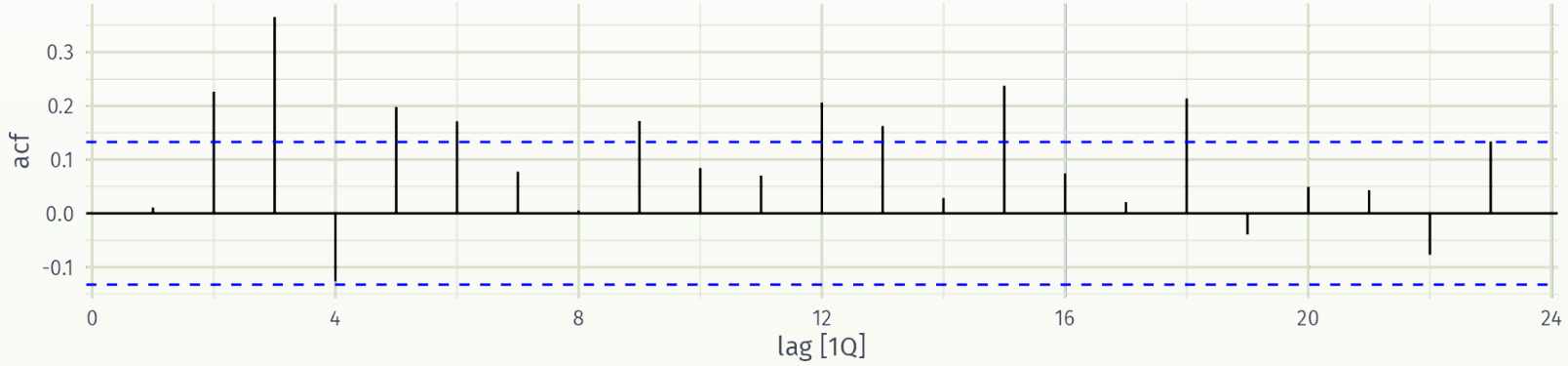
# Beer production - residuals

```
augment(beer_snaive_fit) |>
  autoplot(.resid) +
  labs(x = "Quarter", y = "", title = "Residuals from snaïve method")
```



# Beer production - residual ACF

```
augment(beer_snaive_fit) |> ACF(.resid) |> autoplot()
```




These residuals look like random noise

Strongly disagree Disagree Neutral Agree Strongly agree

0 responses

poll.nectric.com.au/s/africast-modelling

**0** responses

CLOSED 5 connected